

Designing a Healthcare AI Governance Framework

Scenario

Imagine this: You've just stepped into the role of Chief AI Governance Officer at a multinational healthcare technology company. Congrats! The organization operates across 15 countries, leveraging AI for:

Medical diagnosis

Treatment recommendations

Drug discovery

Patient care optimization

Recent global incidents involving unsafe or biased AI use in healthcare have placed your company under increased scrutiny. Your CEO has asked you to design a comprehensive governance framework that ensures your AI systems are safe, ethical, compliant, and accountable—across diverse jurisdictions and use cases.

Your Mission

Design an end-to-end AI governance framework tailored for a multinational healthcare setting that addresses real-world risk, multi-jurisdictional compliance, and organizational accountability.

Assume:

- You have general knowledge of healthcare AI risks and practices.
- You do not need to cite specific country regulations but should consider variation in global data privacy, medical safety, and ethical standards.
- You can define the scope of AI systems based on the categories listed (diagnosis, treatment, drug discovery, care optimization).

Your Tasks

1. Risk Assessment and Categorization (~8 minutes)

Conduct a comprehensive risk scan across the company's AI systems.
Include the following:

Risk categories across technical, operational, ethical, regulatory, and reputational domains

Specific healthcare risks (e.g., misdiagnosis, treatment bias, privacy breaches, clinical safety)

Cross-border complexity (e.g., data localization, consent rules, algorithm transparency)

Risk prioritization based on likelihood and impact

Tip: Focus on how these risks might vary across use cases (diagnosis vs. drug discovery) or regions (EU vs. US).

2. Governance Framework Design (~10 minutes)

Design a scalable and enforceable framework to manage those risks. Include the following:

Organizational structure: Who owns what? (e.g., Data Scientists, Medical Officers, Regulatory Teams)

Decision processes: How are AI systems approved, updated, or suspended?

Monitoring and oversight: What tools or reviews are in place to catch problems early?

Accountability mechanisms: Who is responsible for system outcomes and escalations?

Compliance integration: How will you track obligations across global jurisdictions?

Framework evolution: How will this framework stay up to date as technology and regulations change?

3. Implementation Strategy (~7 minutes)

Lay out a practical plan to bring this framework to life. Include the following:

→ Phased rollout: Which regions or systems get priority?

→ Resources and training: What does your team need to succeed?

→ Success metrics: How will you measure adoption, compliance, and effectiveness?

→ Timeline: Key milestones over the first 6–12 months

Challenges and mitigations: Where might you hit resistance, and how will you address it?

Completion Criteria

Your submission should include:

→ A comprehensive risk assessment, with healthcare-specific and cross-jurisdictional awareness

→ A governance framework with clear roles, structures, and processes

→ A realistic implementation strategy grounded in organizational change principles

→ Logical connections between your three sections (risks → framework → implementation)

Demonstrated understanding of the complexity of healthcare AI governance

Format: Submit your response as a structured outline, slide deck, or visual framework—whatever best communicates your thinking.

After submission, you'll engage in a guided Coursera Coach conversation where you'll test the scalability of your framework

Explore how it would function under pressure (e.g., system failure, compliance audit)

Consider how different teams (legal, medical, engineering) might interact with your model

This is your opportunity to stress-test your ideas and refine your strategy before applying them in real-world or simulated environments.